AI SALES AGENT

User stories

Principal actor goals and Given/When/Then acceptance criteria, traced to the functional requirements.

Phase 1 MVP · 21 October 2026 · Prepared from the canonical documents in docs/

How to read the stories

Each story uses the form required by the product specification and is transcribed from docs/00_product/02_USER_STORIES. Priority matches the requirements register. Staff means any signed-in member unless a role is named. An AI reply in an acceptance criterion assumes auto_send unless the criterion says otherwise. A new business starts in approval_required.

Blanket rules that apply to every story: tenant data never crosses businesses; playground and test orders are excluded from KPIs and billing; destructive actions are confirmed; and an error says what happened and what to do next.

Epic A

Account, workspace, team, and onboarding. Primary actors: Owner and Admin.

US-A-01 — Register an account

As a prospective merchant, I want to create an account with my email and password, so that I can start setting up my AI sales agent.

Priority: Must · Requirements: FR-AUTH-001, FR-AUTH-002, FR-AUTH-010

- Given I am on the registration page, when I submit a valid email, a password meeting the policy, and my name, then an account is created and I am signed in.
- Given an account already exists with that email in any letter case, when I submit, then I see "an account with this email already exists" and no second account is created.
- Given I submit a password below the minimum policy, when I submit, then I see the specific unmet rule and no account is created.
- Given any registration, when I inspect the stored record, then the password appears only as a hash, and the plaintext appears in no log or response body.

US-A-02 — Sign in and stay signed in

As a returning user, I want to sign in and remain signed in across page reloads, so that I do not re-authenticate constantly while working the inbox.

Priority: Must · Requirements: FR-AUTH-003, FR-AUTH-004, FR-AUTH-009

- Given valid credentials, when I sign in, then I receive an access token and a refresh token and land on the dashboard.
- Given my access token has expired but my refresh token is valid, when I make an API call, then the client silently refreshes and the call succeeds without a visible interruption.
- Given invalid credentials, when I attempt sign-in, then I see a generic failure message that does not reveal whether the email exists.
- Given repeated failed attempts beyond the threshold for one email or IP, when I try again, then I am rate-limited with a retry-after indication.

US-A-03 — Sign out

As a user on a shared device, I want to sign out, so that nobody else can use my session.

Priority: Must · Requirements: FR-AUTH-005

- Given I am signed in, when I sign out, then my refresh token is revoked server-side and reusing it returns unauthorised.
- Given I signed out, when I press the browser back button, then protected pages do not render cached tenant data.

US-A-04 — Create my business workspace

As a new user, I want to create my business, so that all my products, conversations, and settings live in one workspace.

Priority: Must · Requirements: FR-BIZ-001, FR-BIZ-002, FR-BIZ-003

- Given I have no business, when I sign in, then I am routed to business creation rather than an empty dashboard.
- Given I submit business name, industry, country, default language, and timezone, then a business is created with me as Owner and currency defaulting to TND for Tunisia.
- Given the business is created, then the system has seeded one default AI agent, default AI settings, references to platform default prompts, and starter delivery/payment/returns rules flagged as unconfirmed defaults.
- Given the business is created, when I open the dashboard, then the onboarding checklist is visible with "business created" complete and the remaining steps listed.

US-A-05 — Invite a teammate

As an owner, I want to invite staff with a role, so that they can work the inbox without gaining control of billing or settings they should not touch.

Priority: Must · Requirements: FR-BIZ-010, FR-BIZ-014, FR-AUTH-011

- Given I am Owner or Admin, when I invite an email with role Admin or Employee, then an invitation is created and the invitee receives a link.
- Given I hold an invitation link, when I accept it (registering if needed), then I join that business with exactly the role specified.
- Given my plan seat limit is reached, when I try to invite, then I see a quota message with an upgrade action and no invitation is created.
- Given I am an Employee, when I open the team page, then I cannot invite or remove members and the controls are absent, not merely disabled.

US-A-06 — Remove a teammate

As an owner, I want to remove a member immediately, so that a departing employee loses access to customer data at once.

Priority: Must · Requirements: FR-BIZ-011

- Given a member exists, when I remove them, then their next API request for that business is rejected.
- Given the removed member had active conversation assignments, then those assignments are released and the conversations return to the unassigned queue.
- Given the removal, then an audit log entry records actor, target, and timestamp.

US-A-07 — Resume onboarding where I left off

As a busy merchant, I want onboarding to remember my progress, so that interruptions do not force me to start over.

Priority: Must · Requirements: FR-BIZ-020, FR-BIZ-021, FR-BIZ-022, FR-BIZ-023

- Given I completed two steps and closed the browser, when I sign in on a different device, then the same two steps show as complete.
- Given a step is optional, when I skip it, then I can still proceed and can return to it from Settings.
- Given blocking steps remain, when I view the checklist, then each unmet item is listed with a direct link to the screen that resolves it.

US-A-08 — Test the AI before customers see it

As an owner, I want to chat with my AI in a playground, so that I can hear its voice and check its facts before it talks to real customers.

Priority: Must · Requirements: FR-BIZ-026, FR-AI-030, FR-AI-031

- Given no channel is connected, when I open the playground, then I can send a message and receive an AI reply using my real catalog and rules.
- Given the playground, when I run a canned scenario (price, angry customer, buying intent, or recommendation), then I see the AI response and the tools called.
- Given I am Owner or Admin, then I can see the decision / reasoning trace.
- Given I am an Employee, then I cannot open the playground (ai.playground is 403); I never see traces.
- Given the playground conversation, then its messages and any orders are marked as test data and are excluded from analytics KPIs.
- Given I complete at least one playground exchange, then the "AI test" onboarding gate is satisfied.

US-A-09 — Activate the AI, but only when it is safe

As an owner, I want the system to stop me activating a half-configured AI, so that my customers never receive a broken first impression.

Priority: Must · Requirements: FR-BIZ-024, FR-BIZ-025 · Canonical gates: 01_PRODUCT_REQUIREMENTS.md §14

- Given one or more activation gates are unmet, when I view the activation screen, then the activation control is disabled and every unmet gate is listed with a link.
- Given gates are unmet, when I call the activation API directly, then it returns a validation error enumerating the specific unmet gates, not a generic rejection.
- Given all five gates are met, when I activate, then AI processing begins on connected channels and I am redirected to the Inbox.
- Given I activate, then the business records who activated and when in the audit log.

US-A-10 — Acknowledge how handoff works

As an owner, I want to be told explicitly that the AI creates pending orders for humans to confirm, so that I do not leave paying customers waiting because I did not know.

Priority: Must · Requirements: FR-HAND-001, FR-BIZ-020

- Given I reach the activation step, when I have not acknowledged the handoff workflow, then I am shown a concise explanation of purchase intent → pending order → human confirmation and must confirm understanding.
- Given I acknowledge, then the gate is satisfied and the acknowledgement is recorded with timestamp and user.

Epic B

Catalog and business knowledge. Primary actors: Owner and Admin.

US-B-01 — Add a product with variants

As an admin, I want to add a product with size and colour variants that each carry their own price and stock, so that the AI answers availability questions accurately.

Priority: Must · Requirements: FR-CAT-001, FR-CAT-004, FR-CAT-005, FR-CAT-006, FR-CAT-007, FR-CAT-013

- Given I create a product with name, description, and category, when I add variants with attributes, price, and stock, then each variant is stored as an independently sellable unit.
- Given I create a product with no variants, then the system creates one default variant carrying the price and stock.
- Given I enter a negative price or negative stock, then the save is rejected with a field-level error.
- Given I set a discount price greater than or equal to the price, then the save is rejected.
- Given I reuse an existing SKU within the same business, then the save is rejected as duplicate.
- Given the AI searches the catalog, then it matches variants (size, color, SKU, price, stock), not an unpriced parent product.

US-B-02 — Upload product images

As an admin, I want to attach images to a product or to a specific variant, so that the inbox and future customer-facing surfaces show the right picture.

Priority: Must · Requirements: FR-CAT-008, FR-CAT-009

- Given I upload an image without selecting a variant, then it is stored as a product-level image.
- Given I upload an image and select the Black variant, then it is associated with that variant only.
- Given multiple images exist in one scope, then exactly one can be primary and setting a new primary clears the previous.
- Given any upload, then the binary is stored in object storage and the database holds only the URL and metadata.

US-B-03 — Import a catalog by CSV

As an admin with 200 products, I want to upload a spreadsheet, so that I do not enter products one by one.

Priority: Must · Requirements: FR-CAT-020 → FR-CAT-027

- Given I download the template, then it contains the required headers (name, price, stock) plus optional columns and one example row.
- Given I upload a file with errors in rows 5 and 12, then I see a row-level report naming the column and reason for each, and nothing has been written to the catalog.
- Given a valid file, when I review the preview, then I see counts of products, variants, and categories to be created or updated, and import runs only after I confirm.
- Given I confirm, then the import runs asynchronously, I can navigate away, and the run's status is visible with row counts and a downloadable error file.
- Given the import completes, then embeddings are enqueued for the affected records and the catalog onboarding step is marked complete.
- Given I upload the identical file a second time, then existing variants are updated by SKU and no duplicates are created.

US-B-04 — Keep the AI's knowledge fresh after edits

As an admin, I want the AI to use my new price the moment I change it, so that I never sell at last week's price.

Priority: Must · Requirements: FR-CAT-011, FR-KNOW-012, FR-KNOW-014

- Given I change a variant price, when a customer asks the price on the next turn, then the reply uses the new price sourced from the database, not from a cached embedding.
- Given I edit a product description, then a re-embedding job is enqueued and the indexing state is visible in the UI.
- Given I trigger a full reindex, then all knowledge documents and product records for the business are re-embedded and progress is reported.

US-B-05 — Define delivery zones and fees

As an owner, I want to state my delivery fees per region, so that the AI quotes shipping correctly instead of guessing.

Priority: Must · Requirements: FR-KNOW-001, FR-KNOW-009

- Given I define Tunis at 5 TND and other regions at 8 TND, when a customer says they are in Sfax, then the AI quotes 8 TND.
- Given a customer names a region I have not defined, then the AI states it must confirm and offers handoff rather than inventing a fee.
- Given any delivery answer, then the quoted figure traces to a business rule record in the decision trace.

US-B-06 — Define payment methods and return policy

As an owner, I want to record which payments I accept and my return window, so that the AI answers consistently and never over-promises.

Priority: Must · Requirements: FR-KNOW-002, FR-KNOW-003, FR-KNOW-004

- Given I accept cash on delivery only, when a customer asks about card payment, then the AI states the accepted methods without offering card.
- Given I set a 7-day return window, when a customer asks about returns, then the AI states 7 days with my conditions.
- Given a customer asks a refund question, then the AI does not decide the refund and escalates per the handoff rules.

US-B-07 — Prevent the AI from discounting

As an owner, I want the AI to have no discount authority by default, so that my margins are not negotiated away automatically.

Priority: Must · Requirements: FR-KNOW-005, FR-AI-021

- Given no discount rule exists, when a customer asks for a lower price, then the AI declines politely and escalates with the negotiation reason.
- Given I define an explicit promotion rule with a valid date range, then the AI may state exactly that promotion and nothing beyond it.
- Given an expired promotion, then the AI does not mention it.

US-B-08 — Handle contradictory rules safely

As an owner who has set conflicting policies, I want the AI to stop rather than pick one, so that it does not commit me to terms I did not intend.

Priority: Must · Requirements: FR-KNOW-010

- Given one enabled rule says free delivery above 100 TND and another says delivery is always 5 TND, when a customer with a 120 TND cart asks about delivery, then the AI does not answer the fee and escalates with reason policy_conflict at normal priority.
- Given the escalation, then the staff briefing names both conflicting rules.

US-B-09 — Add FAQs and policy documents

As an admin, I want to paste my FAQs, so that the AI answers common questions in my own words.

Priority: Must · Requirements: FR-KNOW-011, FR-KNOW-012, FR-KNOW-013

- Given I save a document with a title and body, then it is chunked and embedded asynchronously and its status becomes indexed.
- Given a question semantically matching that document, when a customer asks it, then the reply draws on the document and the retrieval is recorded in the AI action log.
- Given I re-save an unchanged document, then no new embedding work is performed.

Epic C

Channel connection. Primary actors: Owner and Admin. The end customer is affected, but does not log in.

US-C-01 — Connect Instagram and Messenger

As an owner, I want to connect my Facebook Page and Instagram account in a guided flow, so that the AI can start receiving my DMs.

Priority: Must · Requirements: FR-INT-001, FR-INT-002, FR-INT-003, FR-INT-004

- Given I start the Meta connection, then I complete OAuth and am shown the Pages I administer.
- Given I select a Page with a linked Instagram professional account, then both surfaces are created as channel accounts in active status.
- Given the connection succeeds, then credentials are stored encrypted and no endpoint returns the raw token.
- Given I lack the required Meta permissions, then I see which permission is missing and how to grant it, not a raw provider error.
- Given the WhatsApp card on the integrations page, then it shows "Coming in Phase 2" and is not connectable in MVP.

US-C-02 — See channel health at a glance

As an admin, I want to see whether each channel is working, so that I notice a broken connection before customers do.

Priority: Must · Requirements: FR-INT-005, FR-INT-012

- Given a connected account, then the integrations page shows status, last inbound message time, and last error if any.
- Given the provider token expires or permissions are revoked, then the account moves to an error state and Owner and Admin receive a notification.
- Given an account is in error, then the AI does not attempt outbound sends on it and the failure reason is visible.

US-C-03 — Disconnect a channel without losing history

As an owner, I want to disconnect a channel, so that I can stop automation while keeping my conversation and order records.

Priority: Must · Requirements: FR-INT-006

- Given I disconnect a channel account, then inbound processing and outbound sending stop for it immediately.
- Given the disconnection, then existing conversations, messages, customers, and orders remain readable.
- Given I reconnect the same account later, then it links to the existing records rather than creating duplicates.

US-C-04 — Never lose or duplicate an inbound message

As an owner, I want every customer message to appear exactly once, so that I can trust the inbox as a record.

Priority: Must · Requirements: FR-INT-007, FR-INT-008, FR-INT-009, FR-CONV-021

- Given the provider delivers a webhook, then it is signature-verified, persisted raw, and acknowledged within the latency target before any AI work.
- Given the provider re-delivers the same event, then no second message row, no second AI run, and no second order are created.
- Given a webhook with an invalid signature, then it is rejected and logged as a security event.
- Given two messages from the same customer arrive within one second, then replies are not interleaved and only coherent output is sent.

US-C-05 — Receive image messages

As an employee, I want to see photos a customer sends, so that I can understand requests the AI cannot interpret.

Priority: Must · Requirements: FR-INT-013

- Given a customer sends an image with no text, then the image is stored and rendered in the inbox.
- Given an image-only message, then the AI asks the customer to describe or name the product rather than guessing its content.

US-C-06 — WhatsApp service window and templates (Phase 2)

Deferred. See FR-INT-023 → FR-INT-029 and Brain Dump # Integration Architecture → Phase 2. Not tested in MVP.

Priority: — · Requirements: —

US-C-07 — One conversation view regardless of channel

As an employee, I want Instagram and Messenger threads to behave identically, so that I do not learn different tools per channel.

Priority: Must · Requirements: FR-INT-010, FR-CONV-035, FR-CONV-036

- Given conversations from connected channels, then each is rendered with the same layout, actions, and AI controls, differing only by a channel badge.
- Given a message from any channel, then it is stored in the canonical shape and no provider-specific field reaches the inbox or AI.
- Given the AI replies, then the reply is delivered back through the channel the customer used.

Epic D

Inbox, conversations, and AI autonomy. Primary actors: Employee, Admin, and the AI agent.

US-D-01 — Work a prioritised inbox

As an employee, I want conversations needing me to appear at the top, so that I handle the most valuable and most urgent first.

Priority: Must · Requirements: FR-CONV-001, FR-CONV-002, FR-CONV-003, FR-CONV-004

- Given conversations in mixed states, then those in HUMAN_REQUIRED sort above others, ordered by priority critical → high → normal, then oldest handoff first.
- Given I filter by status, channel, or assignee, then the list updates accordingly and the filter persists across navigation within the session.
- Given each row, then it shows customer, channel, last message preview, unread state, status badge, priority badge, and assignee.

US-D-02 — See new activity without refreshing

As an employee, I want the inbox to update live, so that I do not miss a waiting customer.

Priority: Must · Requirements: FR-CONV-005, FR-NOTIF-006

- Given the inbox is open, when a new inbound message arrives, then the conversation appears or moves without a manual refresh.
- Given a conversation escalates while I am viewing the list, then its status and priority badges update in place.
- Given the connection drops, then the UI indicates stale data and recovers automatically when connectivity returns.

US-D-03 — Read the full context of a conversation

As an employee taking over, I want all context on one screen, so that I do not ask the customer to repeat themselves.

Priority: Must · Requirements: FR-CONV-010, FR-CONV-008

- Given I open a conversation, then I see the transcript with sender types visually distinguished, the customer profile, lifecycle stage, order timeline, AI status, and the handoff briefing when present.
- Given long history, then older messages load on demand without losing scroll position.

US-D-04 — Understand why the AI escalated

As an employee, I want a plain answer to "why did the AI transfer this?", so that I know what to do first.

Priority: Must · Requirements: FR-CONV-020, FR-OBS-002, FR-HAND-003

- Given a conversation in HUMAN_REQUIRED, then the UI states the escalation reason, priority, and the required action from the briefing.
- Given I am Owner or Admin and I expand the detail, then I see the recorded decision trace: intent, confidence, rules checked, and outcome.
- Given I am an Employee, then I see the briefing, not the raw reasoning trace.
- Given a purchase handoff, then the briefing shows the pending order id and amount.

US-D-05 — Read a summary instead of the whole thread

As an employee, I want an automatic summary at handoff, so that I can act in seconds rather than reading fifty messages.

Priority: Must · Requirements: FR-CONV-019

- Given a conversation enters HUMAN_REQUIRED, then a summary is generated and stored covering what the customer wants, their concern, what the AI already collected, the handoff reason, products discussed, and the AI's recommendation.
- Given more than ten new messages arrive before a human opens the thread, then the summary is regenerated.
- Given summary generation fails, then a template-based fallback summary is stored rather than nothing.

US-D-06 — Take over from the AI

As an employee, I want to take control instantly, so that the AI does not talk over me.

Priority: Must · Requirements: FR-CONV-011, FR-CONV-012, FR-HAND-004

- Given an AI-active conversation, when I click Take over, then AI replies are disabled for that conversation and the status becomes HUMAN_HANDLING if it was HUMAN_REQUIRED.
- Given I simply send a message without clicking Take over, then the AI is disabled for that conversation automatically.
- Given the AI had a job in flight when I took over, then that job does not send a message after my takeover.

US-D-07 — Give the conversation back to the AI

As an employee, I want to resume AI handling once I have resolved the issue, so that routine follow-ups do not stay on my plate.

Priority: Must · Requirements: FR-CONV-013

- Given no order is awaiting confirmation on this conversation, when I click Resume AI, then AI handling re-enables and the status returns to AI_ACTIVE.
- Given an order on this conversation is in PENDING_CONFIRMATION, when I attempt Resume AI, then it is blocked with an explanation that the order must be confirmed or cancelled first.

US-D-08 — Assign work to a specific person

As an admin, I want to assign conversations, so that two people do not reply to the same customer.

Priority: Must · Requirements: FR-CONV-014

- Given I assign a conversation to a member, then the assignee is shown in the list and detail, and the assignment is recorded with actor and timestamp.
- Given a conversation is reassigned, then the previous assignment is closed and the history is retained.
- Given I filter by "assigned to me", then only my active assignments appear.

US-D-09 — Leave internal notes

As an employee, I want to write notes only my team can see, so that the next person has context without messaging the customer.

Priority: Must · Requirements: FR-CONV-015

- Given I add a note, then it appears in the thread visually distinct from messages and is never delivered to the customer.
- Given any note, then its author and timestamp are shown.

US-D-10 — Approve AI replies before they send

As a cautious owner in my first week, I want to approve each AI reply, so that I build trust before letting it send on its own.

Priority: Must · Requirements: FR-AI-010, FR-AI-011, FR-AI-012, FR-AI-015

- Given my business is new, then the reply mode defaults to approval_required.
- Given the AI generates a reply in approval_required, then it is stored as a draft, is not delivered, and appears in an actionable queue.
- Given I approve a draft, then it is delivered to the customer and recorded as an AI-sent message.
- Given I edit a draft before approving, then the edited text is delivered and the edit is recorded for quality analysis.
- Given I switch to auto_send, then subsequent replies deliver without approval.

US-D-11 — Configure the AI's voice

As an owner, I want to set my agent's name, tone, and language, so that it sounds like my brand and not a generic robot.

Priority: Must · Requirements: FR-AI-002, FR-AI-003, FR-AI-004, FR-AI-007

- Given I set agent name, personality, tone, default language, and greeting, then the next AI reply reflects them without a redeploy.
- Given I set the default language to Arabic, then replies default to Arabic while still matching a customer who writes in French.
- Given I adjust the confidence threshold, then escalation behaviour changes on the next turn.

US-D-12 — Trust that the AI never invents facts

As an owner, I want guaranteed grounding, so that the AI never quotes a price or stock level that is not real.

Priority: Must · Requirements: FR-AI-020, FR-AI-021, FR-AI-023

- Given a customer asks the price of a product, then the figure in the reply matches the current variant price exactly and the tool call is logged for that turn.
- Given a product does not exist in my catalog, then the AI says it is unavailable and offers alternatives or handoff, and never describes a fabricated item.
- Given a stock check returns zero, then the AI does not offer the item as purchasable.
- Given tools return conflicting data, then the AI states uncertainty and offers handoff instead of choosing.

US-D-13 — Handle a customer the AI does not understand

As an owner, I want the AI to ask once and then get help, so that confused customers are not trapped in a loop.

Priority: Must · Requirements: FR-AI-024

- Given classifier confidence is below the threshold, then the AI asks exactly one clarifying question.
- Given the following turn is also below the threshold, then the conversation escalates with reason low_confidence at normal priority.
- Given escalation, then the AI sends no further autonomous replies on that conversation.

US-D-14 — Keep working when the AI provider fails

As an owner, I want the product to degrade gracefully, so that an outage at a model vendor does not silently lose my customers.

Priority: Must · Requirements: FR-AI-028, FR-AI-029, NFR-REL-07

- Given the primary model errors, then the system retries with the fallback model and the message is answered normally.
- Given all model attempts fail or exceed the latency ceiling, then the conversation escalates to human with an integration or timeout reason and I am notified.
- Given an outage, then the inbox remains usable and humans can still send messages.
- Given an outage, then no degraded or empty AI message is delivered to the customer.

US-D-15 — Close a conversation

As an employee, I want to close resolved conversations, so that my queue reflects real work.

Priority: Must · Requirements: FR-CONV-016, FR-CONV-017

- Given I close a conversation, then its status becomes RESOLVED and it leaves the active queue.
- Given a closed conversation receives a new inbound message, then it reopens into the appropriate active state.
- Given a conversation has no activity for the configured window and no pending order, then it auto-closes and the transition is logged as system-initiated.

US-D-16 — Know who matters without opening anything

As an owner with 200 conversations today, I want the list ordered by who is worth my attention, so that I never scroll looking for the buyer.

Priority: Must · Requirements: FR-CONV-031, FR-CONV-032, FR-CONV-033, FR-CONV-034, FR-LEAD-009

- Given conversations in mixed states, then those needing a human appear above all AI-handled ones regardless of lead score.
- Given two conversations both needing a human, then the higher handoff priority sorts first, and equal priorities sort oldest-waiting first.
- Given AI-handled conversations, then they sort by temperature, then score, then recency.
- Given an angry customer scoring 20 and a hot lead scoring 95, then the angry customer appears first.
- Given I open the inbox for the first time in a session, then the default view is "Needs me", not "All".
- Given a list card, then it shows customer, message snippet, temperature badge, product of interest, value, channel, assignee, and relative time without my opening it.

US-D-17 — Work every channel in one place

As an employee, I want one inbox for Instagram and Messenger, so that I am not switching apps to find out who is waiting.

Priority: Must · Requirements: FR-CONV-035, FR-CONV-036

- Given conversations from multiple channels, then they appear in a single list with a channel icon per row.
- Given I filter by channel, then the list narrows without leaving the unified view.
- Given I have not connected a given channel, then it does not appear as a filter option at all.
- Given the product, then there is no per-channel page anywhere in the navigation.

US-D-18 — Get the customer's context beside the chat

As an employee, I want identity, lead state, and history next to the conversation, so that I can act without opening three screens.

Priority: Must · Requirements: FR-CONV-030, FR-CONV-039, FR-CONV-040, FR-CONV-041, FR-CONV-042

- Given an open conversation, then the right pane shows name, phone, product of interest, lead score, plain-language status, and a timeline of what has happened.
- Given the timeline, then it merges messages, state changes, AI actions, and order events in chronological order.
- Given the panel, then no more than four primary actions are offered and they match the conversation's current state.
- Given a stored phone number, then the call action opens a telephone link rather than requiring me to copy the number.

US-D-19 — Trust the lead score

As an owner, I want to see why a lead is hot, so that I believe the ordering instead of ignoring it.

Priority: Must · Requirements: FR-LEAD-001, FR-LEAD-003, FR-LEAD-007, FR-LEAD-002

- Given a scored conversation, then I can reveal the contributing signals and their point values.
- Given the score is displayed, then it is never labelled as a probability or likelihood of purchase.
- Given a purchase intent is detected, then the score rises and the temperature badge updates without a refresh.
- Given a conversation is resolved or its order cancelled, then the score resets.
- Given scoring runs, then no LLM call is made for it.

US-D-20 — Let the AI help while I am the one replying

As an employee handling a conversation, I want a suggested reply I can send or edit, so that I keep the AI's speed while keeping control.

Priority: Must · Requirements: FR-AI-016, FR-AI-017, FR-AI-018, FR-AI-019, FR-AI-020a, FR-AI-020b

- Given I am handling a conversation, when I request a suggestion, then a grounded draft appears with send, edit, and dismiss actions.
- Given a suggestion is displayed during HUMAN_REQUIRED, then nothing is delivered to the customer until I act — the paused-AI guarantee still holds.
- Given the conversation is a complaint or refund, then no suggestion is offered.
- Given I send a suggestion, then it is recorded as an AI message flagged as human-approved.
- Given I edit before sending, then the edited text is sent and the edit is captured for quality analysis.
- Given suggestions are generated, then each generation is recorded in AI usage and is not triggered on every keystroke.

US-D-21 — See my sales as a pipeline

As an owner, I want a Kanban view of leads and orders, so that I can see where revenue is stuck.

Priority: Must · Requirements: FR-CONV-050, FR-CONV-051, FR-CONV-052, FR-CONV-053

- Given the pipeline page, then I see four columns — new leads, qualified, payment pending, sold — each with a count and total value.
- Given a card, then it shows customer, product, value, and temperature, and opens the conversation when clicked.
- Given the payment-pending column, then card copy states the order is awaiting my confirmation, never awaiting customer payment.
- Given the pipeline, then it introduces no status that does not already exist on a conversation or an order.

US-D-22 — Not be spammed

As an owner, I want to be notified only when something needs me, so that I keep notifications turned on.

Priority: Must · Requirements: FR-NOTIF-010, FR-NOTIF-011, FR-NOTIF-012, FR-NOTIF-013, FR-NOTIF-014

- Given the AI is handling conversations normally, then I receive no notifications at all.
- Given a hot lead, an order request, AI uncertainty, or an angry customer, then I am notified.
- Given the same conversation triggers the same type repeatedly, then I receive at most one notification per hour for it.
- Given several hot leads within a short window, then they arrive as one grouped notification.
- Given a non-critical notification, then it accumulates in the notification centre rather than interrupting me.

US-D-23 — Keep a correct screen when the connection drops

As an employee on a weak mobile connection, I want the inbox to stay accurate, so that I do not miss a customer because a socket died.

Priority: Must · Requirements: FR-RT-001, FR-RT-003, FR-RT-004, FR-RT-005

- Given an open inbox, when a message arrives, then it appears within the realtime latency target without a refresh.
- Given my connection drops, then the UI indicates the degraded state and continues updating by polling.
- Given my connection is restored, then the client refetches the list and open thread so no missed event leaves the screen wrong.
- Given any pushed update, then the same data is retrievable through the REST API.

Epic E

Purchase handoff and orders. Primary actors: the AI agent, Employee, and Owner.

US-E-01 — AI detects buying intent and collects details

As an owner, I want the AI to gather order information when someone wants to buy, so that I receive a ready-to-confirm order instead of a raw chat.

Priority: Must · Requirements: FR-HAND-008, FR-CUST-003

- Given a customer expresses intent to buy, then the AI confirms the specific product and variant before collecting anything else.
- Given collection begins, then the AI asks for one piece of information per turn (name, phone, address, quantity), not all at once.
- Given the customer provides information, then it is validated and written to the customer record.
- Given the customer supplies an unparseable phone number, then the AI asks again once with a format hint before escalating.
- Given collection is in progress, then the conversation status is ORDER_INTENT_DETECTED.

US-E-02 — AI reserves stock during collection

As an owner, I want the last unit held while details are collected, so that I do not sell the same item to two customers.

Priority: Must · Requirements: FR-ORD-010, FR-ORD-011, FR-ORD-012

- Given the AI confirms a variant with buying intent, then a reservation is created for the requested quantity with a 30-minute expiry.
- Given a reservation is active, then available stock reported to other conversations excludes it.
- Given the order is created, then the reservation is marked committed.
- Given collection is abandoned or the reservation expires, then it is released by the sweep job and stock becomes available again.
- Given only one unit exists and two conversations reach intent simultaneously, then exactly one reservation succeeds and the other customer is offered alternatives.

US-E-03 — AI creates a pending order and hands off

As an owner, I want a real order record the moment the customer is ready, so that nothing depends on someone reading the chat.

Priority: Must · Requirements: FR-ORD-001, FR-ORD-002, FR-HAND-002, FR-HAND-003, FR-HAND-004

- Given all required fields are collected, then the AI re-verifies stock and creates an order with status PENDING_CONFIRMATION and payment status NOT_COMPLETED.
- Given the order is created, then it stores the customer snapshot, line items with unit prices, total, currency, source conversation, and AI as creator.
- Given the order is created, then the AI triggers handoff with reason purchase_handoff at high priority, the conversation becomes HUMAN_REQUIRED, and AI replies are disabled.
- Given handoff, then a briefing is persisted containing identity, summary, products, reason, order id, amount, and required action.
- Given handoff, then the AI tells the customer a team member will confirm, and makes no promise about payment method, timing, or discount.

US-E-04 — Stock disappears mid-conversation

As an owner, I want the AI to handle a sold-out item gracefully, so that I never have to cancel an order I could not fulfil.

Priority: Must · Requirements: FR-HAND-009

- Given stock fell to zero between the first check and order creation, then no order is created.
- Given this situation, then the AI apologises, states the item is unavailable, and offers in-stock alternatives.

US-E-05 — Staff are notified of a new order request

As an owner, I want an unmissable alert when the AI produces an order, so that I confirm it while the customer is still warm.

Priority: Must · Requirements: FR-NOTIF-001, FR-NOTIF-002, FR-NOTIF-003, FR-NOTIF-004, FR-HAND-005

- Given a pending order is created, then an in-app notification is created for Owner and Admin (and the assignee if set) showing customer, product, amount, and waiting status.
- Given the notification, then clicking it opens the source conversation with the order in context.
- Given multiple notifications, then they sort by priority then recency and the unread count is accurate.
- Given I mark one or all as read, then the count updates and the state persists across sessions.

US-E-06 — Confirm an order in one action

As an employee, I want to confirm the order from the conversation, so that I do not navigate to a separate orders screen mid-chat.

Priority: Must · Requirements: FR-ORD-004, FR-ORD-005, FR-HAND-010, FR-HAND-006

- Given a pending order on the open conversation, then a confirm action is visible with the order summary.
- Given I confirm, then the order becomes CONFIRMED, order_confirmed_at is recorded, and the conversation advances to ORDER_CONFIRMED.
- Given a human first opens or takes the handed-off conversation, then handoff_completed_at is recorded once.
- Given I confirm, then the conversion KPI reflects it in the next rollup.

US-E-07 — Cancel a pending order

As an employee, I want to cancel an order the customer no longer wants, so that my stock and metrics stay accurate.

Priority: Must · Requirements: FR-ORD-004, FR-ORD-008

- Given a pending order, when I cancel it with a reason, then its status becomes CANCELLED and the associated reservation is released.
- Given cancellation, then the conversation can be resumed by the AI if the staff member chooses.
- Given an invalid transition (for example CANCELLED back to PENDING_CONFIRMATION), then the API rejects it.

US-E-08 — Record offline payment

As an employee collecting cash on delivery, I want to mark payment received, so that the record reflects reality.

Priority: Must · Requirements: FR-ORD-006

- Given a confirmed order, when I mark payment completed, then the payment status updates and the actor and timestamp are recorded.
- Given payment status changes, then it is independent of order status; an order may be confirmed and unpaid.

US-E-09 — Create an order manually

As an employee handling an unusual request, I want to create an order myself, so that the system holds all sales, not only AI-generated ones.

Priority: Must · Requirements: FR-ORD-003, FR-ORD-009

- Given I create an order manually with customer, items, and quantities, then it is saved with the human creator recorded.
- Given the orders list, then I can distinguish AI-created from manually created orders.
- Given manual orders, then they are excluded from the AI-attributed conversion metric.

US-E-10 — Escalate non-purchase situations correctly

As an owner, I want every sensitive situation routed to a human with the right urgency, so that the AI never mishandles an angry or legal matter.

Priority: Must · Requirements: FR-HAND-001, FR-HAND-002, FR-HAND-007 · Canonical: 06_AI_HUMAN_HANDOFF_RULES.md

- Given a customer expresses anger or dissatisfaction, then the conversation escalates at critical priority with the complaint reason and the AI attempts no upsell.
- Given a refund request, then the AI never approves or denies it and escalates at critical priority.
- Given a customer asks for a human, then the AI escalates at high priority and confirms politely that someone will come.
- Given a legally sensitive topic, then the AI escalates at critical priority without offering an opinion.
- Given a VIP-tagged customer and the VIP auto-handoff flag enabled, then the conversation escalates at critical priority.
- Given a bulk or out-of-catalog request, then the AI escalates at high priority with the request captured in the briefing.
- Given any escalation, then the reason and priority are recorded and the notification reflects the priority.

Epic F

Customers and notifications. Primary actors: Employee and Admin.

US-F-01 — Recognise a returning customer

As an owner, I want the AI to know who it is talking to, so that repeat buyers feel remembered instead of interrogated.

Priority: Must · Requirements: FR-CUST-001, FR-CUST-006, FR-CUST-007, FR-CUST-011

- Given a customer messages from a channel identity seen before, then the existing customer record is matched, not duplicated.
- Given a new Instagram or Messenger identity shares a verified phone with an existing customer, then the identity is auto-linked to that customer.
- Given only a similar name or an unverified phone, then a separate customer is created (no auto-merge).
- Given a customer with a confirmed past order, then their lifecycle stage is at least customer and the AI may reference the prior purchase.
- Given a customer in order_pending, then the AI provides status updates only and does not attempt to sell again.
- Given a brand-new identity, then a customer record is created at unknown_visitor.

US-F-02 — Maintain a customer profile

As an employee, I want to view and correct customer details, so that deliveries reach the right place.

Priority: Must · Requirements: FR-CUST-002, FR-CUST-004, FR-CUST-005

- Given a customer record, then I can view name, phone, email, address, language, tags, notes, conversations, and orders in one place.
- Given I edit a field, then the change is saved and audited.
- Given I add a tag or note, then it is visible to the team and to the AI context where applicable.

US-F-03 — Flag a VIP

As an owner, I want to mark my best customers, so that they always get a human quickly.

Priority: Must · Requirements: FR-CUST-008

- Given I tag a customer VIP, then their lifecycle stage reflects VIP treatment.
- Given VIP auto-handoff is enabled for my business, then their next conversation escalates at critical priority.
- Given VIP auto-handoff is disabled, then the AI applies the VIP tone rules without forcing escalation.

US-F-04 — Export customer data

As an owner, I want to export my customers, so that my data is not locked in.

Priority: Should · Requirements: FR-CUST-009

- Given I request an export, then I receive a file containing my business's customers only.
- Given I am an Employee, then the export action is unavailable.
- Given an export occurs, then it is recorded in the audit log.

US-F-05 — Merge duplicate customers

As an admin, I want to merge two customer records that are the same person on different channels, so that history is not split.

Priority: Must · Requirements: FR-CUST-010, FR-CUST-011

- Given two customers on my business, when I merge them as Owner or Admin, then identities, orders, notes, and conversations attach to the surviving record and the merge is audited.
- Given I am an Employee, then merge is unavailable.
- Given only similar names, then the system does not merge automatically.

Epic G

Analytics, billing, observability, and platform controls. Primary actors: Owner, Admin, and the platform operator.

US-G-01 — See whether the AI is making me money

As an owner, I want four clear numbers, so that I can decide whether to keep paying.

Priority: Must · Requirements: FR-ANL-001, FR-ANL-002, FR-ANL-003 · Canonical definitions: 01_PRODUCT_REQUIREMENTS.md §15

- Given a selected period, then I see messages handled, orders created, human handoffs, and conversion percentage.
- Given handoffs, then I can see the breakdown by escalation reason.
- Given I reload, then the same period returns identical figures.
- Given my business timezone, then period boundaries follow it rather than server time.
- Given sandbox or test data exists, then it is excluded from all four figures.

US-G-02 — Understand my usage against my plan

As an owner, I want to see how many messages I have used, so that I am not surprised when the AI stops.

Priority: Must · Requirements: FR-BILL-004, FR-BILL-005, FR-BILL-006, FR-BILL-010

- Given the usage page, then I see current period consumption against my plan limits for messages, AI replies, conversations, channels, seats, and monthly AI budget when set.
- Given I cross 80% and 95% of a hard limit, then I receive a warning surfaced in the dashboard.

US-G-03 — Hit my limit without losing customers

As an owner over my plan limit, I want messages to keep arriving, so that I can still serve customers manually while I decide to upgrade.

Priority: Must · Requirements: FR-BILL-007, FR-BILL-008

- Given I reach any hard limit (messages, AI replies, conversations, or AI budget), then AI auto-replies stop.
- Given the limit is reached, then inbound messages continue to be received and displayed and humans can still reply.
- Given the limit is reached, then the owner is notified and an upgrade call to action is visible in the dashboard and the inbox.
- Given I am at the channel limit, then connecting another channel is blocked with a clear explanation.

US-G-04 — Rate the AI's answers

As an employee, I want to thumb-rate AI messages, so that the agent improves instead of repeating mistakes.

Priority: Must · Requirements: FR-OBS-006, FR-OBS-007

- Given any AI message, then I can rate it positive or negative.
- Given a negative rating, then I must select a reason from the fixed enum and may add a comment.
- Given feedback exists, then Owner and Admin can filter it by date, reason, and prompt version.
- Given a rating, then the prompt version of the rated message is stored with it.

US-G-05 — Audit what the AI did

As an admin debugging a bad interaction, I want to see every tool call, so that I can tell whether the data or the model was wrong.

Priority: Must · Requirements: FR-OBS-001, FR-OBS-008

- Given a conversation, then I can view its AI action log with tool name, input, output, success, error, and duration per call.
- Given a rate-limit event, then it appears in the action log.
- Given a failed tool call, then the error is recorded and the conversation behaviour matches the uncertainty rule.

US-G-06 — Watch AI cost

As an owner, I want to see what the AI costs me, so that I understand my margins.

Priority: Should · Requirements: FR-OBS-004, FR-OBS-005

- Given AI turns occurred, then token counts and estimated cost are recorded per turn with model and prompt version.
- Given the usage summary, then Owner sees cost aggregated by day and model.
- Given I am an Employee, then no monetary AI data is visible to me.

US-G-07 — Roll out features per tenant

As a platform operator, I want to enable a capability for one business, so that I can pilot without deploying.

Priority: Must · Requirements: FR-FLAG-001, FR-FLAG-002, FR-FLAG-003, FR-FLAG-004

- Given I enable a flag for a business, then the behaviour changes on the next request with no redeploy.
- Given a platform-level default, then businesses without an explicit override inherit it.
- Given a disabled channel flag, then that channel is not offered in the integrations UI.

US-G-08 — Version prompts safely

As a platform operator, I want prompt changes to be new versions, so that I can compare quality and roll back.

Priority: Must · Requirements: FR-AI-026, FR-AI-027, FR-AI-025

- Given I change a prompt, then a new version is created and the previous version remains intact.
- Given only one version per type may be active, then activating a new version deactivates the prior one atomically.
- Given any AI message, then the prompt version used is recorded and retrievable.
- Given a prompt or model change, then the golden set runs in CI and a regression blocks the merge.

US-G-10 — Score AI conversations

As an admin reviewing quality, I want conversation-level evaluations stored, so that we can see where the AI fails.

Priority: Must · Requirements: FR-AI-032, FR-OBS-011

- Given a golden-set run, production sample, or staff review, then an ai_evaluations row is stored with score, failure reason, and evaluator type.
- Given evaluations exist, then they are distinct from per-message thumbs (ai_feedback).

US-G-09 — Prove tenant isolation

As an owner, I want certainty that no other merchant can see my data, so that I can trust the platform with my customers.

Priority: Must · Requirements: FR-BIZ-006, NFR-SEC-04, NFR-SEC-05

- Given a valid token for business A, when I request a resource id belonging to business B, then the response is not found or forbidden and never leaks existence details.
- Given any list endpoint, then results contain only the active business's records.
- Given an Employee calls an Owner-only endpoint directly, then it is rejected server-side regardless of UI state.

Epic H

End-customer experience. The shopper never sees the product UI.

US-H-01 — Get an instant, correct answer

As a shopper messaging a store at 11pm, I want an immediate accurate reply, so that I do not wait until morning to decide.

Priority: Must · Requirements: FR-AI-020, NFR-PERF-02

- Given I ask about a product, then I receive a reply within the latency target with the real price and availability.
- Given the store is closed, then I still get an answer and, where a human is needed, a realistic callback expectation based on working hours.

US-H-02 — Be understood in my own language

As a Tunisian shopper who mixes Arabic and French, I want to be answered naturally, so that the conversation does not feel like a form.

Priority: Must · Requirements: FR-AI-022, NFR-I18N-01

- Given I write in Tunisian Arabic, then the reply is in Tunisian Arabic.
- Given I code-switch within one message, then the reply is coherent and in the dominant language of my message.
- Given I switch languages mid-conversation, then the AI follows my switch.

US-H-03 — Buy without friction

As a shopper who has decided, I want to be asked only what is necessary, so that ordering feels like a conversation, not paperwork.

Priority: Must · Requirements: FR-HAND-008

- Given I say I want to buy, then I am asked one question at a time and never re-asked for information I already gave.
- Given I finish giving details, then I am told clearly that a team member will confirm my order.
- Given the handoff, then I am not promised a payment link, discount, or delivery date that was not defined by the merchant.

US-H-04 — Reach a human when I need one

As an upset shopper, I want a person, so that I am not stuck arguing with software.

Priority: Must · Requirements: FR-HAND-001, FR-HAND-007

- Given I ask for a human, then the AI stops replying autonomously and tells me a team member will respond.
- Given I express anger, then no upsell or sales language appears in the final AI message before handoff.
- Given I request a refund, then the AI does not decide it and defers to a human.